



7. Physical and Mechanical Testing

Performance testing of materials such as rubber and rubber-based products, plastic and plastic products, construction materials and school supplies based on ASTM, ISO and BPS PNS methods.

Office or Division:	Standards and Testing Division	
Classification:	Highly Technical	
Type of Transaction:	G2C, G2B, G2G	
Who may avail:	All	
CHECKLIST OF REQUIREMENTS		WHERE TO SECURE
1. Valid sample(s) for testing (refer to the Schedule of Fees and Charges for Testing and Analytical Services)	Provided by the customer	
2. Accomplished Customer Information Sheet (CIS), GP 4.4-01-F01 (1 original)	ITDI-STD /Receiving and Releasing Unit (RRU) or through email correspondence	
3. Valid identification cards - Student ID/Senior Citizen ID/ PWD ID (1 original)	Provided by the customer	
4. Validated Technical Service Request (TSR), GP 4.4-01-F03 (1 original)	Validation Room/ITDI-STD	
5. Invoice receipt (1 original)	ITDI-Administrative Division/2 nd Floor Metrology Building, Cashier Section	
6. Authorization letter if representative (1 original copy)	Provided by the customer	
7. Alternative to authorization letter: accomplished "Authority to Claim" at the back portion of TSR (1 original)	Validation Room/ITDI-STD	



8. Valid government-issued identification cards of the person authorized and the person who gave the authorization (1 photocopy)		Provided by the customer		
9. Client Satisfaction Measurement (CSM) Form, QMS-F2 (1 original/softcopy)		ITDI-STD / Receiving and Releasing Unit (RRU) or through email correspondence		
CLIENT STEPS	AGENCY ACTION	FEES TO BE PAID	PROCESSING TIME	PERSON RESPONSIBLE
1. Customer logs in on the logbook of the guard-on-duty.	1.1 Guard-on-duty endorses customer to the Receiving and Releasing Unit (RRU).	None	1 minute	<i>Guard-on-duty</i> STD Building
2. Customer informs RRU staff of the testing needs and brings sample for submission.	2.1 RRU checks if customer requirement is within the capability of ITDI-STD. If capability is not within ITDI-STD capability, RRU staff endorses customer to OneLab Referral System through the Customer Receiving Officer (CRO)	None	1 minute	<i>Science Research Analyst</i> STD-RRU
3. Customer accomplishes Customer Information Sheet (CIS, GP 4.4-01-F01) or presents the accomplished CIS given through email: pptl.std@itdi.dost.gov.ph	3.1 If within capability, RRU instructs customer to accomplish Customer Information Sheet (CIS, GP 4.4-01-F01). RRU staff calls validator from concerned laboratory through paging system or local phone call. RRU staff	None	2 minutes	<i>Science Research Analyst</i> STD-RRU



	instructs customer to proceed to the Validation Room/designated area.			
4. Customer proceeds to validation room/designated area and wait for the laboratory/section validator.	4.1 Concerned laboratory/ section validator proceeds to Validation Room.	None	3 minutes	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
	4.2 Validator checks validity of sample and completeness of details provided by the customer.	None	3 minutes	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
	4.3 Validator inputs information provided by the customer to the Unified Laboratory Information Management System (ULIMS).	None	10 minutes*	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
5. Customer checks correctness of information in the printed TSR.	5.1 Validator prints 1 copy of the Technical Service Request Form (TSR, GP 4.4-01-F03).	None	1 minute	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
	5.2 Validator discusses to customer due date of testing service, terms and conditions, modes of release of test reports and the requirements for release of test reports to authorized representative.	None	3 minutes	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
6. Customer and validator agree to the content of	6.1 Once agreed by the customer, validator prints	None	2 minutes	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room



the TSR and both sign the form.	three (3) more copies of the TSR.			
	6.2 Customer and validator agree to the content of the TSR and both sign the form, all four (4) copies.	None	3 minutes	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
	6.3 Validator gives three (3) copies of the signed four (4) copies to the customer for payment at the Cashier.	None	1 minute	<i>Supervising SRS</i> Physical and Performance Testing Laboratory Validation Room
7. Customer proceeds to ITDI cashier for payment (option 1); E-payment (option 2) Customer presents three (3) copies of the signed TSRs together with payment of total fee.	7.1 Cashier checks submitted TSRs and issues Invoice of the payment received.	Please refer to ITDI-STD published Schedule of Fees and Charges 20% discount for students, senior citizens, and persons with disabilities (PWDs)	5 minutes	<i>Administrative Officer III</i> Cashier Section, Administrative Division, 2 nd floor Metrology Building
	7.2 Cashier provides the customer his/her copy of the TSR and the invoice. Cashier gives one (1) copy of TSR to Commission on Audit (CoA) and one (1) copy is sent back to ITDI-STD for commencement of	None		<i>Administrative Officer III</i> Cashier Section, Administrative Division, 2 nd floor Metrology Building



	technical service. Laboratory/ section concerned checks payment of customer through online using ULIMS or through received TSR and technical service is started.			
8. No activity.	8.1 Laboratory conducts analytical testing.	None	19 days* (paused clock)	<i>Supervising SRS</i> Physical and Performance Testing Laboratory
9. Customer goes back to ITDI-STD to claim Test Report after the agreed turn-around time for the technical service availed. Customer logs in on the logbook of the guard-on-duty.	9.1 Guard-on-duty endorses customer to the Receiving and Releasing Unit (RRU).	None	1 minute	<i>Guard-on-duty</i> STD Building
10. Customer presents his/her copy of the Technical Service Request (TSR, GP 4.4-01-F03). If a representative is sent, authority to claim and valid identification cards are presented.	10.1 RRU staff checks the presented TSR. If person claiming test report is not the same person who submitted the sample, RRU staff checks the authorization to claim (separate authorization letter or accomplished back page of TSR, GP 4.4-01-F03) the Test Report and valid identification.	None	2 minutes	<i>Science Research Analyst</i> STD-RRU



	For customers who availed of email correspondence as the mode of release of Test Reports, RRU staff immediately sends the Test Report to the provided email address by the customer.			
11. Customer signs the receiving copy of the Test Report.	11.1 RRU staff asks customer to sign the receiving copy of the Test Report.	None	1 minute	Science Research Analyst STD-RRU
	11.2 RRU staff places Test Report inside an envelope and forwards to the customer. Electronic copies of test reports can also be issued through email (if requested during sample validation): rru_std@itdi.dost.gov.ph	None	1 minute	Science Research Analyst STD-RRU
12. Customer accomplishes harmonized Client Satisfaction Measurement (CSM) Form (QMS-F2) and drops accomplished form inside the drop box at the RRU.	12.1 RRU staff asks customer to accomplish Client Satisfaction Measurement (CSM) Form (QMS-F2) or through online (along with the electronic copies of test reports)	None	3 minutes	Science Research Analyst STD-RRU



TOTAL:	Please refer to ITDI-STD published Schedule of Fees and Charges	19 days and 43 minutes	
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*Timelines indicated are for 1 sample submitted only. Longer time is expected if many samples are submitted.

**SCHEDULE OF FEES AND CHARGES FOR TESTING
AND ANALYTICAL SERVICES**
(As of October 1, 2014)

Sample Name	Test Name	Test Method	TEST FEE (PhP)	SAMPLE REQUIREMENT
SACKS	Breaking Strength One way	ASTM D5034	600.00	3 pcs whole sack per sample
	Breaking Strength Other way	ASTM D5034	600.00	
	Bursting Strength, One way	ASTM D3786	450	
	Bursting Strength, Other way	ASTM D3786	450	
	Fabric count, One way	ASTM D3775	200	
	Fabric count, Other way	ASTM D3775	200	
DUTY FREE SHOPPING BAGS	Tensile Elongation, One way	ASTM D882	600	20 pcs bags per sample/per size
	Tensile Elongation, Other way	ASTM D882	600	
	Tear Strength	ASTM D1004	600	
	Sealing Strength	ASTM D882	600	
	Handle Performance	ASTM D882	600	
	Thickness	ASTM D2103	200	
	Sampling	ASTM D882	200	



Sample Name	Test Name	Test Method	TEST FEE (Php)	SAMPLE REQUIREMENT
PERLITE INSULATION	Density	ASTM D4052/ASTM D1298	350.00	5 pcs, 200mmx100mmx50mm
	Moisture content	ASTM C610 sec 11.1.7	350.00	5 pcs, 150mmx150mmx50mm
	Conditioning (Oven @24hrs, 23°C, 50%RH)		1000.00	na
	Flexural strength	ASTM C203	600.00	5 pcs, 12"L x 4"W x 1"T
	Compressive strength	ASTM C185	600	5 pcs, 6"L x 4"W x 1/2"T
PVC PIPES	Resistance to acetone	ISO 3472	250	5 pcs, 100mm x 25mm
	Resistance to H2SO4 for 14 days @55°C	ISO 3473	9400	5 pcs, Area = 45±3cu cm
	Longitudinal heat reversion	ISO 2505	350	5 pcs, L = 200mm
	Vicat softening point	ISO2507	1000	5 pcs, Rectangular bar L=50mm, W= 10-20mm, T=2.4 - 6mm
PVC/HDPE/PPR PIPES	Water absorption	ISO 2508	1000	5 pcs, Pipe OD=32mm, Area =50 sq cm OD >32mm or L = 5cm
	Pipe stiffness	ASTM D2412	1000	5 pcs, Pipe L=6 in(150 mm())
	Flattening	ASTM D2241 Sec 8.6	1000	5 pcs, L= 2 in
	Hydrostatic Test	ISO 1167	1500	5 pcs, L is 250 x 3xOD + 100mm
RUBBER PRODUCTS	Aging test	ASTM D573		
	22h		1100	
	48 h		2400	
	70 h		3500	
	72 h		3600	
	96 h		4800	
	100 h		5000	
	168 h		8400	
	336 h		16800	



	Compression Set	ASTM D395	1000	5 pcs, Cylindrical disk type, 1" diameter
	Density/Specific gravity	ASTM D792	450	5 pcs, about 1.5 g each
	Durometer hardness	ASTM D2240/ISO 868	250	10 pcs, flat, t=6mm minimum
	Tensile Strength & Elongation One-way	ASTM D412	600	2 sheets, smooth & flat, 305mm x 305mm , T = 1.3-3.0 mm
	Tensile Strength & Elongation Other way		600	
	Tear resistance, One-way	ASTM D 624	600	2 sheets, smooth & flat, 305mm x 305mm , T = 1.3-3.0 mm
	Tear resistance, Other-way		600	
	Brittleness -Low temperature	ISO 812	2600	2 sheets, smooth & flat, 305mm x 305mm , T = 2.0 mm
	Abrasion resistance	ASTM D4060	1500	5 pcs, 4"(100mm) x 4"(100mm), T=0.125 (3mm) or less"
	Sampling		200	pls see sample size
	Shore Durometer Hardness (Type A)	ASTM D2240-06	250	10 pieces, smooth and flat L = W = 10cm; T ≥ 6mm
	International Rubber Hardness Degree (IRHD) - Method N	ISO 48:2007	1100	5 pieces, smooth and flat L = W = 2in ; T = 8 - 10mm
	International Rubber Hardness Degree (IRHD) - Method M	ISO 48:2007	1100	5 pieces, smooth and flat L = W = 2in ; T = 2mm
	Ozone Resistance - 4 Sample Static	PNS ISO 1431-1	25000	1 sheet, smooth and flat L = W = 305mm (12in) T = 2.0mm
	Ozone Resistance - 3 Sample Dynamic	PNS ISO 1431-1	25000	
	Volume / Surface Resistivity	ISO 14309	1200	7 pieces, smooth and flat L = W = 4 in (100mm) T = 1 - 2mm



Sample Name	Test Name	Test Method	TEST FEE (Php)	SAMPLE REQUIREMENT
PLASTIC PRODUCTS	Chemical immersion test	ASTM D543	350	5 pcs, L=50mm, W=25.4mm
	Compression	ASTM D695	600 1000	7 pcs, smooth & flat, L=2 in(50.8mm)
	Compression for Bar	ASTM D695	600 1000	7 pcs, W=T= 0.50 in(12.7 mm)
	Compression for Cylinder	ASTM D695	600 1000	7 pcs, D = 0.50 in (12.7 mm)
	Flexural	ISO 178/ASTM D790	600 1000	7 pcs, smooth & flat, T=4mm, W=10mm, L=100mm
	Flexural	ISO 178/ASTM D790	600 1000	7 pcs, T>3.2 mm, W< 0.25 x L, L=20 x T
	Flexural	ISO 178/ASTM D790	600 1000	7 pcs, 1.8mm≤T≤3.2mm, W=12.7 mm(1/2in), L=20 x T
	Water Absorption	ASTM D570	350	50mmL x 25.4mm W, % pcs
	Tension Test	ASTM D638 / D882	600 1000	5 pieces dumbbell shape or W= 10 mm to 25.4 mm (15mm Preferably) L=125mm to 250mm
	Heat Deflection Temperature	ASTM D648	1000	5 pieces, smooth flat surface, L = 127mm W =13mm, Thickness = 3 to 13mm
	Vicat Softening Point	ASTM D1525	1000	5 pcs, (L = 50mm, W = 10 to 20mm, t = 2.4 to 6mm) (if t = 2.4 mm, use two layers of at least t = 2.4 total Thickness)
	Izod Impact	ASTM D256	650	10 pcs, L = 63.5 or 60.3mm, t = 3.17 to 3.4mm, W = 12.7mm,
	Charpy Impact	ASTM 179	650	10 pcs, 4.0 mm x 80 mm
	Hardness Test	ASTM 2240	250	2 pieces flat surface 10 x 10 cm, t = 6mm min



	Abrasion Resistance Test	ASTM D4060	2000	4" x 4" 1/8" t or less 5pcs
	Conditioning (at least 40 hrs @ 23 +/- 2° C, 50 +/- 5% RH)		1000	
	Flammability - Horizontal Burning	UL 94	2300	6 Pieces, smooth and flat L = 125mm, W = 13.0mm, T = 3.0mm
	Flammability - Vertical Burning - 50W (20mm)	UL 94	2300	10 Pieces, smooth and flat L = 125mm, W = 13.0mm, T = 2.0mm - 10.0mm
	Flammability - Vertical Burning - 500W (125mm)	UL94	2300	10 Pieces, smooth and flat L = 125mm, W = 13.0mm, T = 3.0mm and 6 Pieces, smooth and flat <i>plaque</i> L = W 150mm, T = in min or ≤ 1.3mm
	Flammability - Vertical Burning - Thin Material	UL94	2300	10 Pieces, smooth and flat L = 200mm, W = 50mm, T = ≤ 0.025mm and in max
	Flammability - Horizontal Burning - Foamed Material	UL94	2300	10 Pieces, smooth and flat L = 150mm, W = 50mm, T = in min and ≤ 13mm
	Brittleness -Low temperature	ISO 812	2600	2 sheets, smooth & flat, 305mm x 305mm , T = 2.0 mm
EPOXY	Compression	ASTM C356	600 1000	2" x 2" x 2", 5pcs
PLASTIC RESIN	Melting Point	ASTM D2117	300	100 grams (granules, pellets or powder)
	Melt Flow Rate	ASTM D1238	1000	100 grams (granules, pellets or power)
	Density	ASTM D792	450	100 grams (granules, pellets or power)



PVC Powder	Bulk Density / Apparent Density	ASTM D1895	250	0.5 kilograms
Particle Board	Density	PHILSA 106 / App. B/D		
	Block		350	5 pcs, 6" x 6"
	Pipe		350	1 pipe section
	Compression		600	5 pcs, 6" x 6"
	Flexural		600	5 pcs, 12" L x 4" W x 2" T
	Water Absorption		350	5 pcs, 6" x 6"
	Linear Shrinkage		400	5 pcs, 6" x 6"
Adhesive / Sealant	Peel strength	ASTM D903	600 1000	10 pcs, 12" L x 1" W overlap by 6"
	Shear strength	ASTM D1002	600 1000	10 pcs, 5" L x 1" W overlap by 1
	Sampling Fee	BS 3261	150	
Vinyl Tiles	Tension Test	ASTM D4060	600 1000	10 pcs, 12" x 12"
	Resistance Test / Chemicals		350	
	Dimension Test		600	
	Size		200	
	Thickness		200	
	Width		200	
	Taber Abrasion		1500	5 pcs, 4" x 4"
Joint Filter	Compression	ASTM D505	2200	2 pcs, 12" x 12"
	Recovery Density			
	Water Absorption			
SCHOOL AND OFFICE SUPPLIES :				
Masking / Scotch / Packaging Tape	Adhesive Strength	PNS 712	1000	3 rolls
Glue	Adhesive Strength	PNS 1800-2006	600	2 jars
Rubber Band	Tensile Strength / Elongation	PNS 1479	600	1 box 100 pcs



	Dimensional Measurement		200	
Ball Point Pens & Refills	Writing Performance	ISO 12757 - 2 PNS 558	400	10 pcs
	Drying Time Test		400	
	Water Resistance		350	
White Board Marker	Writing Performance	PNS 1227	400	15 pcs
	Impact Resistance		600	
	Heat Resistance			
	Shaft Holding Force Point		600	
	Primary Erase Ability		400	
Permanent Pen	Writing Performance	PNS 2049	400	15 pcs
	Impact Resistance		600	
	Shaft Holding Force Point		600	
	Drying Time		400	
	Resistance to Water		400	
Paper Fastener	Bend Test	PNS 235	400	10 pcs
Chalk	Breaking Strength	PNS 1122	600	25 pcs
Pencil	Breaking Strength	PNS 1413	600	10 pcs
Lead for mechanical pencil	Bending strength	PNS 1471-3 /ISO 9177-3	600	12 pcs/1 box
Crayons	Performance test	PNS 1224	600	1 box
Adhesive Tape	Breaking strength	PNS 712/PNS 2041	1000	5 rolls
	Adhesive Strength	PNS 712/PNS 2041	1000	5 rolls
PVC Electrical Tape	Exposure to Heat 113° C (168 h)	PNS 79	8400	10 rolls
	Tension		600	
	Appearance		250	
	Peeling Off		600	
	Tension After heat Aging @ 100° C for 4 hrs.		700	



Crane and other Lifting Equipment	Load Testing, 50 Tons & 200 Tons	ISO 4310	3500	
Burned Refractory Brick	Bulk Density	ASTM C20 - Sec 12	400	3 pcs, 50mm cube
	Water Absorption		350	3 pcs, 76.2mm cube
Concrete Interlocking Roofing Tiles	Transverse	Philsa 242	400	6 pcs full size
	Water Absorption		350	
Concrete Hollow Block	Compression	ASTM C39	1000	5 pcs / size
Concrete Cylinder	Compression		1000	3 pcs, 152.4 mm x 304 mm with capping
Wire Rod	Tension Test	ASTM A370	600	3 pcs, 13" L
	Flexural / Bend Test		600	5 pcs, 12" L
Hydraulic Cement Mortar	Compression	ASTM C109	1000	3 pcs, min 50 mm cube
Cement Powder	Permeability Test / Fineness Cement	ASTM C204	400	1 kg powder cement
	Initial & Final Setting of Cement	ASTM C191	800	